Name: SOLUTIONS

Date: ____________

Year 9 Ace/Maritime/General Mathematics

Investigation 5, 2015

Topic – Transformation of Quadratic Equations

In Class Component

47

Total Time: *60 minutes* **Reading Time:** *5 minutes*

Working Time: *55 minutes*

Equipment: *Take Home Component*
Calculator

AIM: *In this assessment, you will be investigating the transformation of Quadratic Equations.*

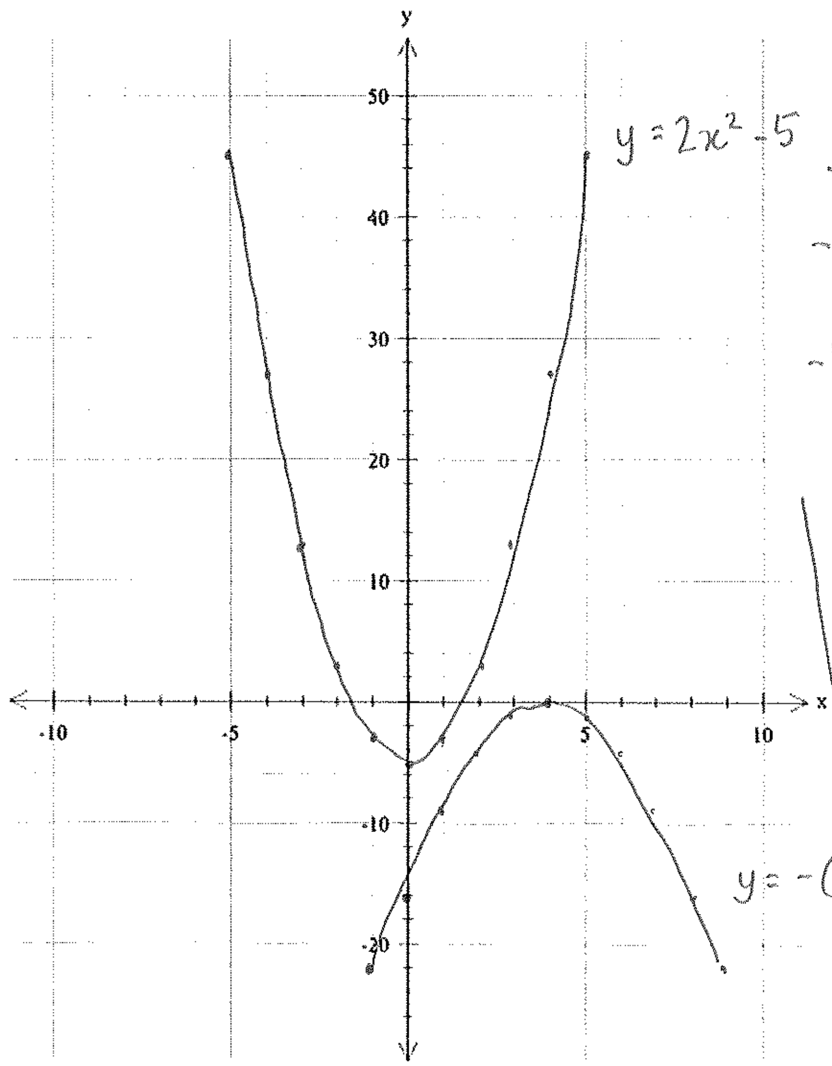
1. [10 marks; 2, 2, 3 3]

a) Complete the table of values for the equations shown.

$x =$	-5	-4	-3	-2	- 1	0	1	2	3	4	5
$y = 2x^2 - 5$	45	27	13	3	-3	-5	-3	3	13	27	45

$x =$	-1	0	1	2	3	4	5	6	7	8	9
$y = -(x-4)^2$	-25	-16	-9	-4	-1	0	-1	-4	-9	-16	-25

b) Draw each of these equations on the axis below, be sure to label each function. $-\frac{1}{2}$ each error



$-\frac{1}{2}$ no label / colour coordinated

$-\frac{1}{2}$ not smooth / straight lines

* follow through only if graph still looks like a quadratic

2. 10 marks; 2, 2, 3, 3]

a) Complete the table of values for the equations shown.

$x =$	0	1	2	3	4	5	6	7	8
$y = \frac{1}{2}(x-4)^2 + 2$	10	6.5	4	2.5	2	2.5	4	6.5	10

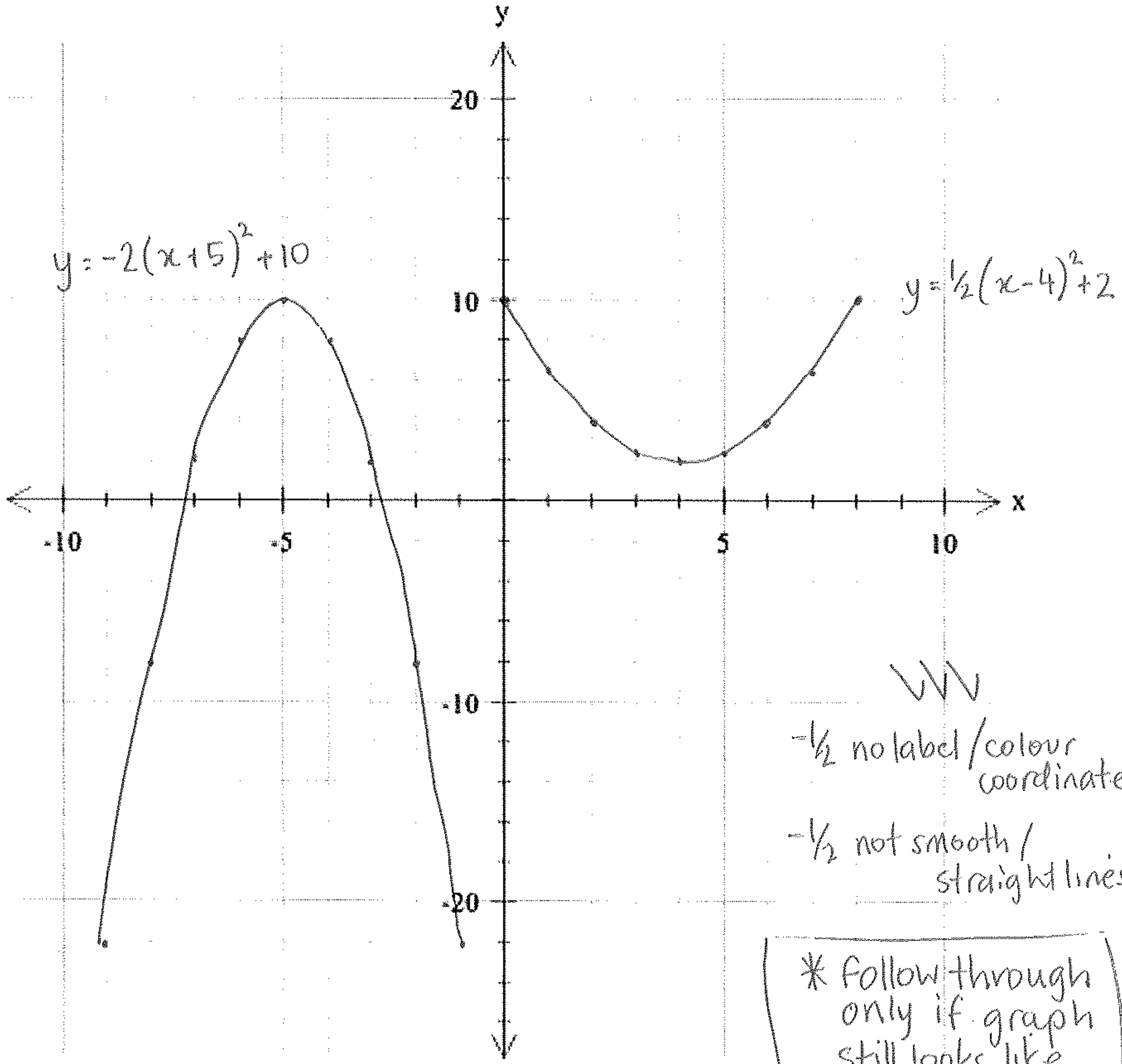
✓✓

$x =$	-9	-8	-7	-6	-5	-4	-3	-2	-1
$y = -2(x+5)^2 + 10$	-22	-8	2	8	10	8	2	-8	-22

✓✓

b) Draw each of these equations on the axis below, be sure to label each function.

$-\frac{1}{2}$ each error



✓✓✓

$-\frac{1}{2}$ no label / colour coordinated

$-\frac{1}{2}$ not smooth / straight lines

* follow through only if graph still looks like a quadratic

Transformation Terminology
Dilation
This is when the graph is thinner (closer to the y-axis) or is wider (closer to the x-axis).
Reflection
This is when the graph has been reflected or flipped over the x-axis.
Vertical Translation
This is when the graph has been moved or translated up or down a specific number of units.
Horizontal Translation
This is when the graph has been moved or translated left or right a specific number of units.

[15 marks; 1, 2, 3, 5, 4]

3. **Using the terminology** above describe how the original graph $y = x^2$ changes the following equations:

*NO marks if NOT correct terminology

a) $y = 4x^2$

Dilated → closer to y-axis
½ ½

b) $y = (x+1)^2$

Horizontal translation 1 unit left
½ ½ ½ ½

c) $y = -x^2 + 13$

Reflection over x-axis
½ ½

Vertical translation 13 units up
½ ½ ½ ½

d) $y = 2(x+9)^2 - 3$

Dilated toward y-axis
½ ½

Horizontal translation 9 units left
½ ½ ½ ½

Vertical translation 3 units down
½ ½ ½ ½

e) $y = -\frac{1}{2}(x-6)^2$

Reflection over x-axis
½ ½

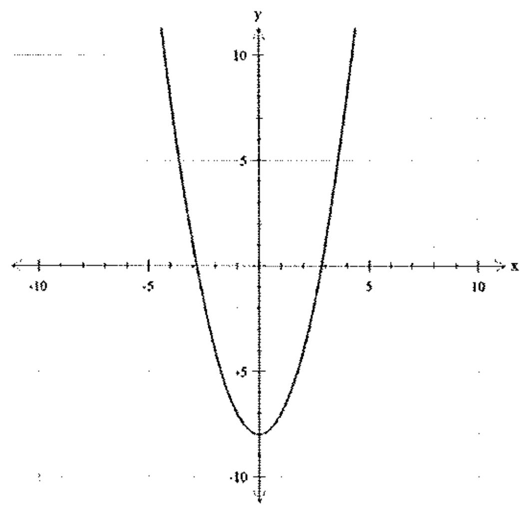
Dilation closer to x-axis / away from y-axis
½ ½

Horizontal translation 6 units right
½ ½ ½ ½

4. [3 marks; 1, 1, 1]

Choose the correct equation that matches each graph:

a)

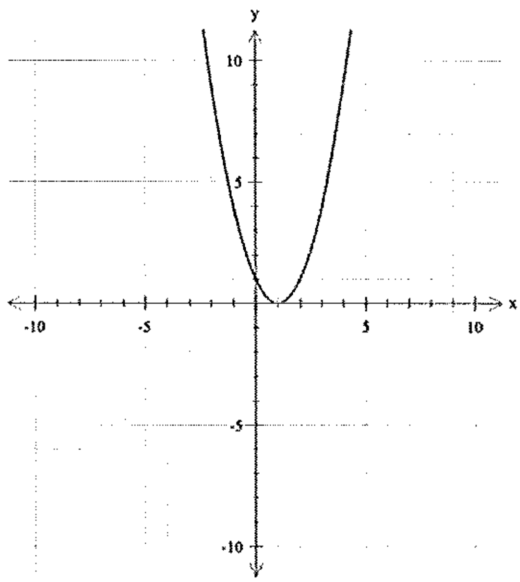


$y = (x-8)^2$

$y = x^2 + 8$

$y = x^2 - 8$

b)

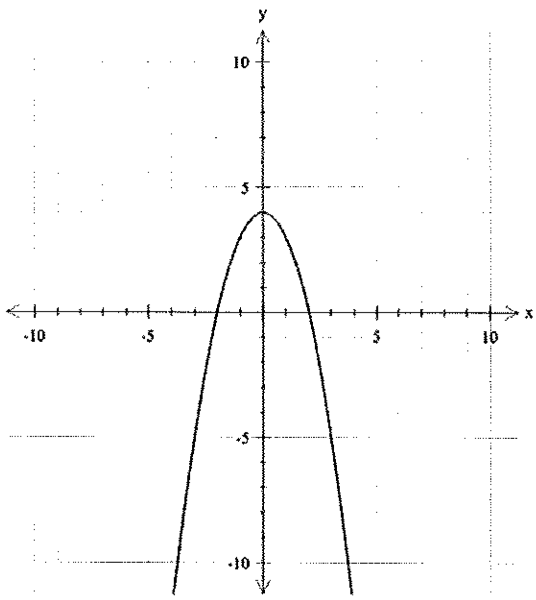


$y = (x+1)^2$

$y = (x-1)^2$

$y = x^2 + 1$

c)



$y = -x^2 + 4$

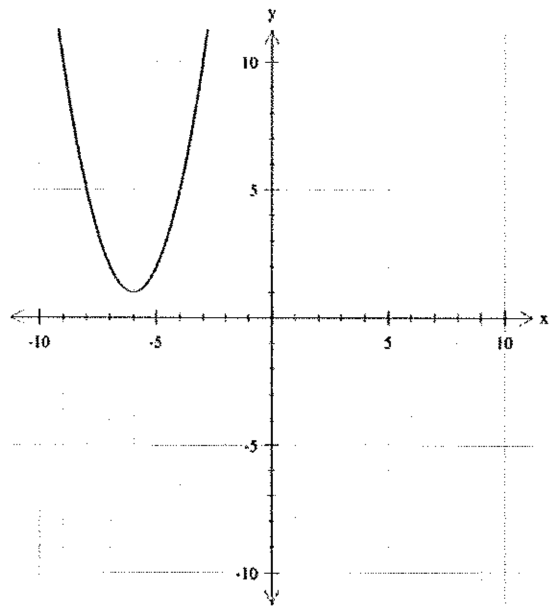
$y = -(x+4)^2$

$y = x^2 + 4$

5. [3 marks; 1, 1, 1]

Choose the correct equation that matches each graph:

a)

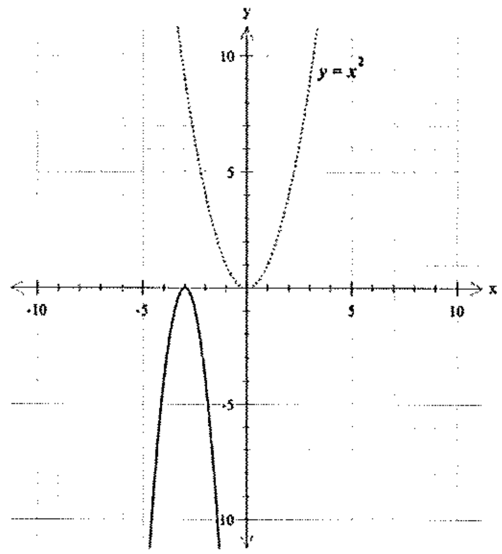


$y=(x-6)^2+1$

$y=(x+6)^2+1$

$y=(x+1)^2-6$

b)

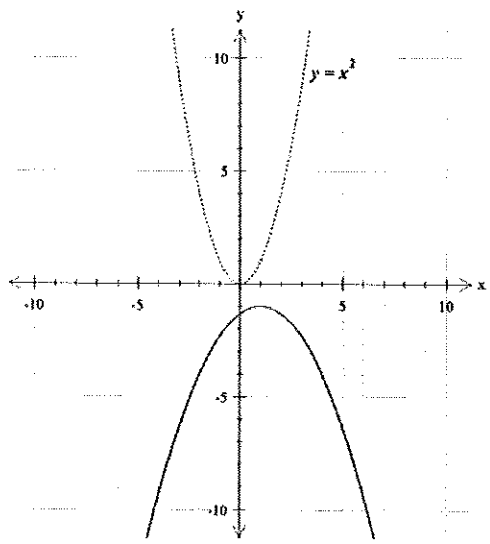


$y=-(x+3)^2$

$y=-4(x+3)^2$

$y=-\frac{1}{4}(x+3)^2$

c)



$y=-3(x+1)^2-1$

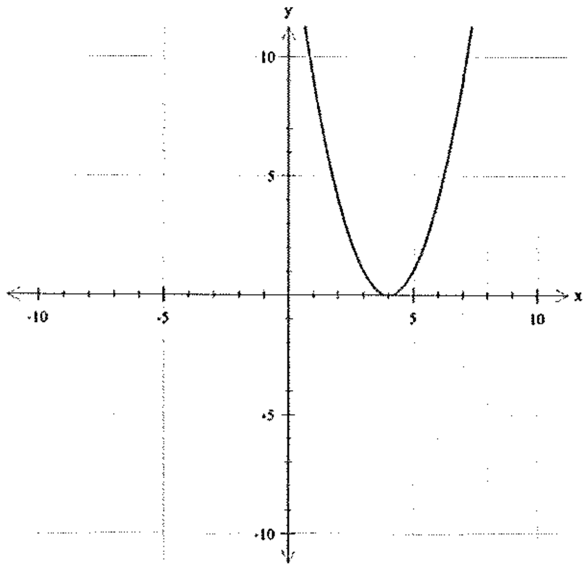
$y=-3(x-1)^2+1$

$y=-\frac{1}{3}(x-1)^2-1$

6. [4 marks; 2, 2]

Using the information you have discovered about transforming quadratics write the equation for each of the following quadratic graphs. Please note there has been **no dilation.**

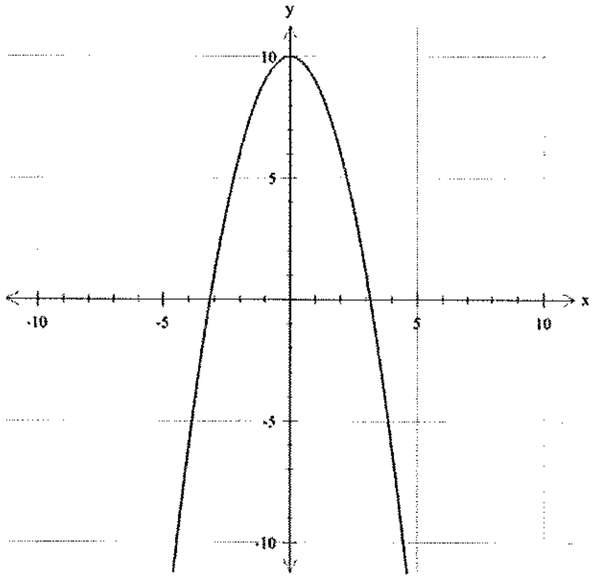
a)



$y = (x-4)^2$

✓✓

b)



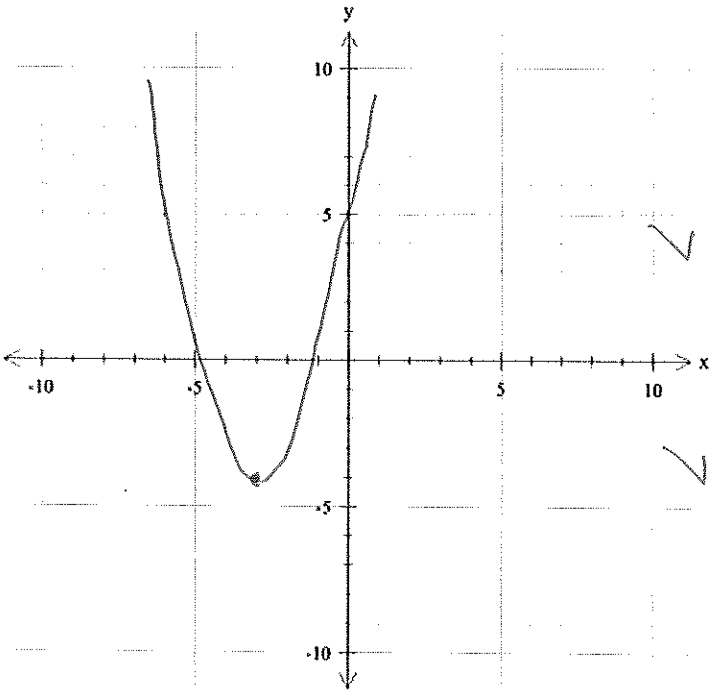
$y = -x^2 + 10$

✓ ✓

7. [2 marks]

Draw a sketch of the following quadratic given the equations below:

$y = (x+3)^2 - 4$



✓ Turning point in correct position $(-3,-4)$

✓ General shape

END OF TEST